Module 04 Lab - Exploratory Data Analysis**Objective:** To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model.**This lab is fully coded.** Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading**What is Exploratory Data Analysis (EDA)?**EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions.We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Seaborn is a library built on top of Matplotlib that makes creating beautiful plots easier.# Load the dataset directly from a
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
print("--- First 5 Rows ---")
print(df.head())
print("--- Basic Info ---")
# .info() is a great first command. It tells us the column names, how many non-null values are in each column, and their data t
```

Show hidden output

Part 2: Descriptive StatisticsLet's start by getting a high-level numerical summary of the data. The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columnsprint("--- Descriptive Statistics ---")
print(df.describe())
print("--- Key Insights from Statistics ---")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].max()}.")
```

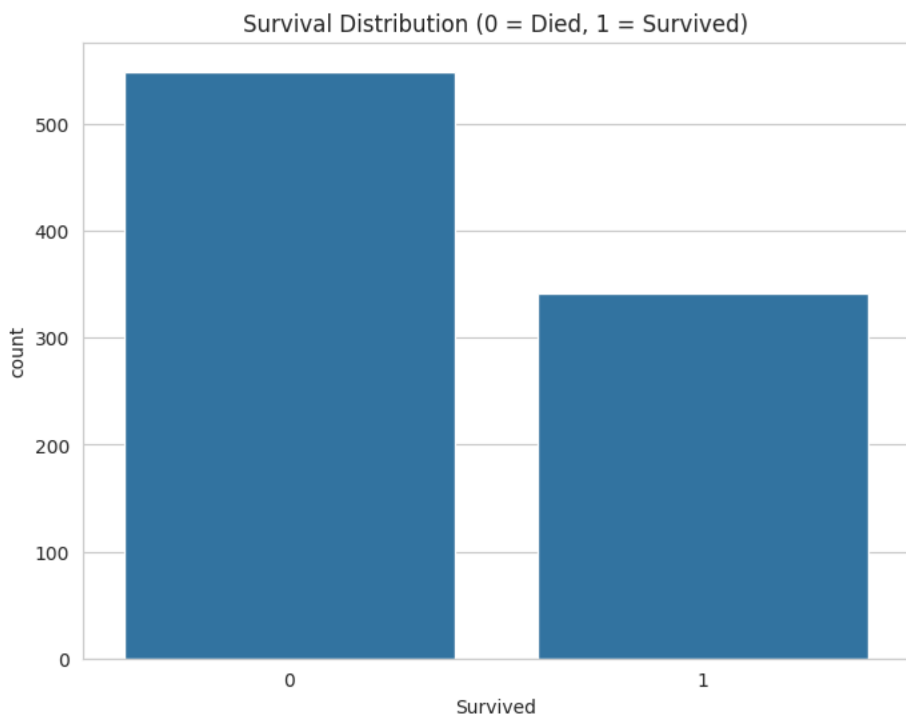
```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
--- Key Insights from Statistics ---
```

```
The average age of a passenger was 29.7 years.
The overall survival rate was 38.4%.
Fares ranged from $0.0 to a whopping $512.3292.
```

Part 3: Visual EDA - Telling Stories with PlotsNumbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights.**A Note on Plotting Libraries:*** **Matplotlib:** The foundational library, gives you full control over everything.* **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived?A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.

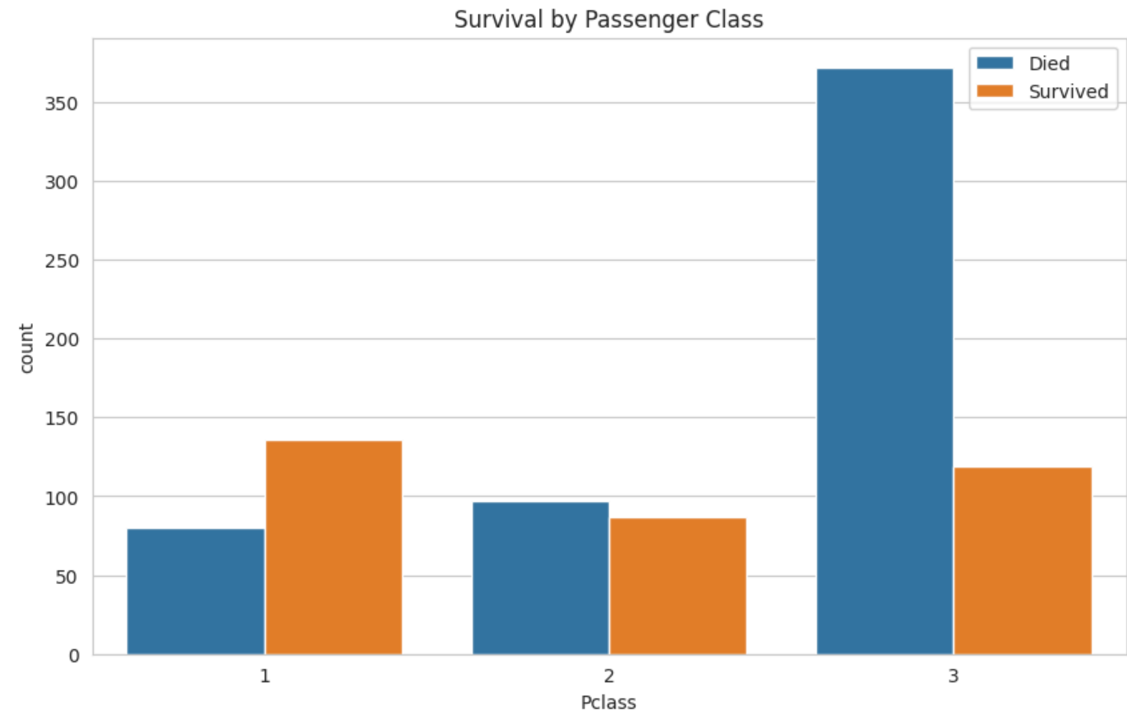
```
sns.set_style('whitegrid') # Sets a nice visual style for our
plt.figure(figsize=(8, 6))
#This shows amount that suriveved
sns.countplot(x='Survived', data=df)
plt.title('Survival Distribution (0 = Died, 1 = Survived)')
plt.show()
print("Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challe
```




```
Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challenge for
```

Visualization 2: Does passenger class matter for survival?Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

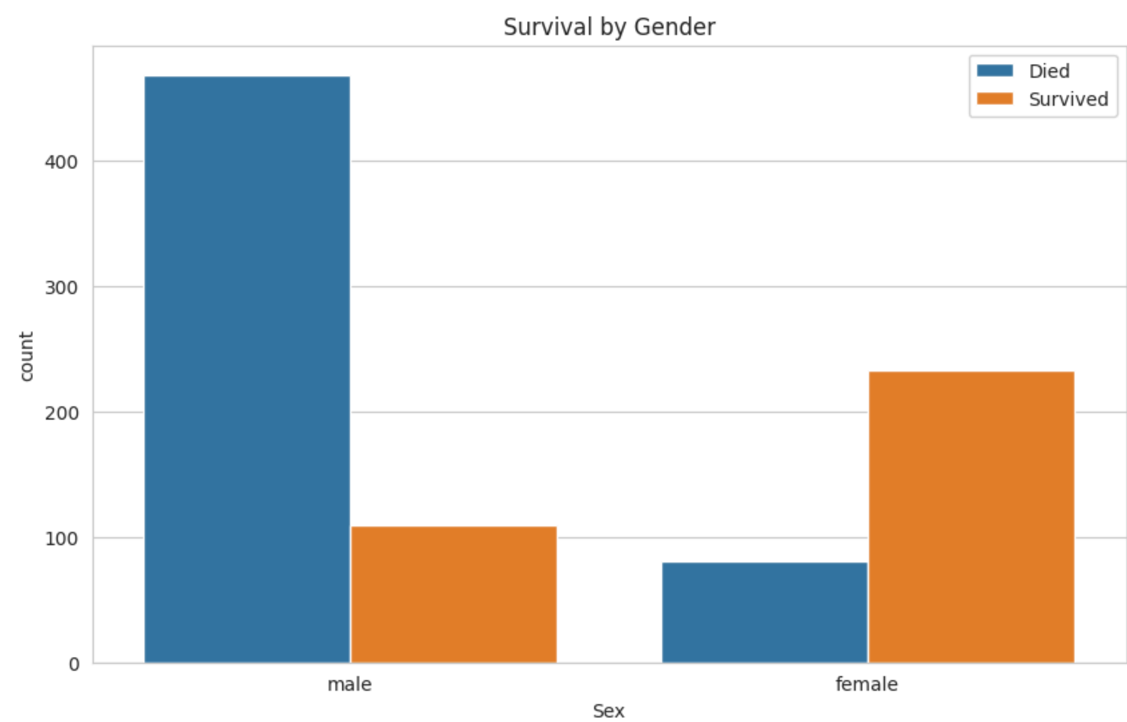
```
plt.figure(figsize=(10, 6))
#Shows the class and shows the survivors from the dataSet
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class pa
```




```
Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passenge
```

Visualization 3: What about gender?The 'women and children first' mantra is famous. Let's see if the data supports it.

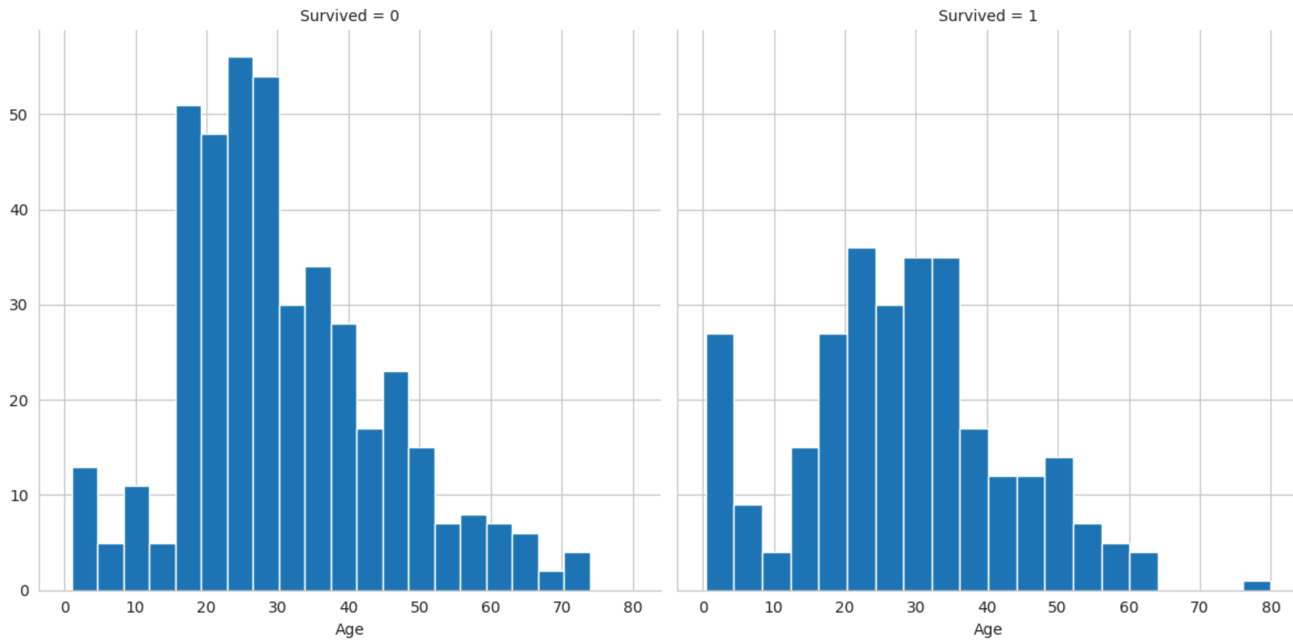
```
plt.figure(figsize=(10, 6))
#Shows the genders that were more or less likey to survive
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very :
```




```
Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong
```

Visualization 4: How does age play a role?For a continuous variable like `Age`, a histogram is a great way to see its distribution.

```
# A FacetGrid allows us to create multiple plots side-by-side to compare distributions.
# Here, we create one histogram for passengers who died (col='Survived'=0) and one for those who survived (col='Survived'=1).
g = sns.FacetGrid(df, col='Survived', height=6)
#Shows a histogram of who survived and who died and there age
g.map(plt.hist, 'Age', bins=20)
plt.show()
print("Insight: The age distribution for those who did not survive is centered around the 20-40 age range. For those who survive
```

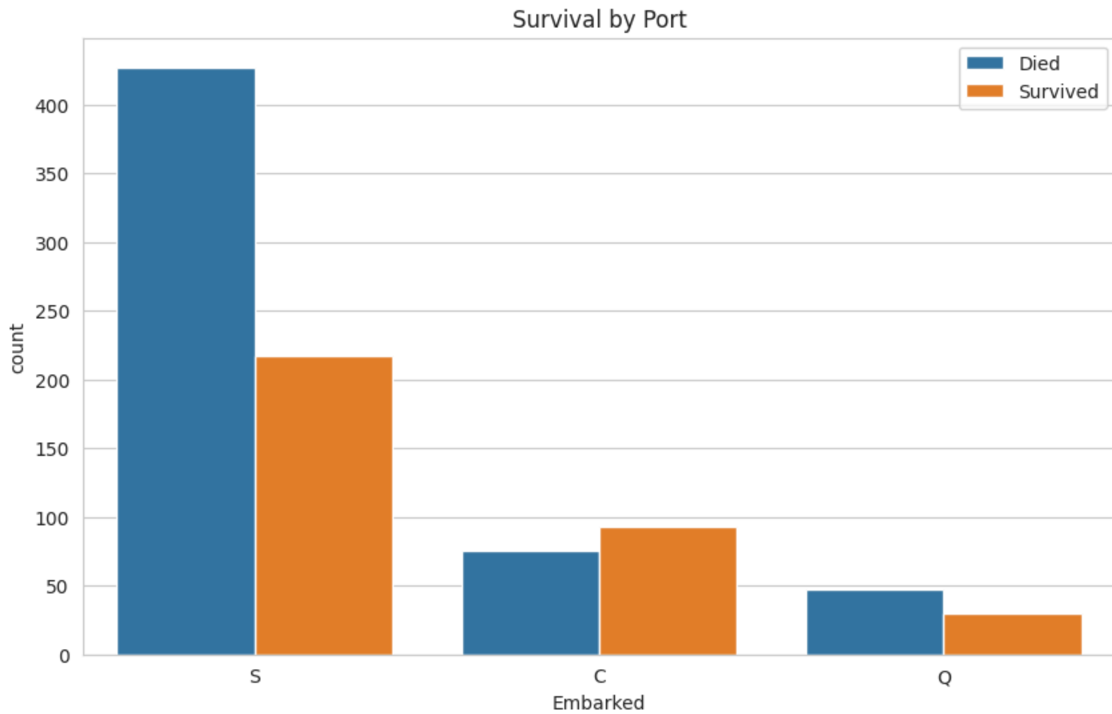



```
Insight: The age distribution for those who did not survive is centered around the 20-40 age range. For those who survived, ther
```

Part 4: Student Experimentation**Instructions:** Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation1. The `Embarked` column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

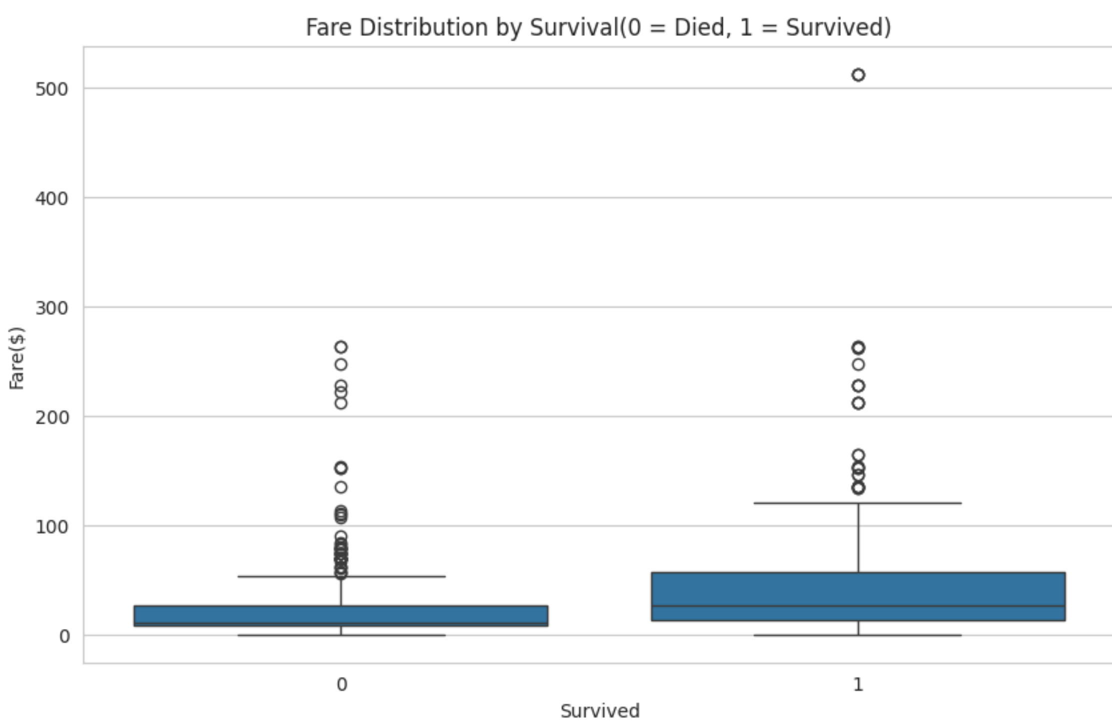
```
plt.figure(figsize=(10,6))
#Shows a countplot where person embarked and if they survived or not
sns.countplot(x = 'Embarked', hue='Survived', data=df)
plt.title('Survival by Port')
plt.legend(['Died','Survived'])
plt.show()
print("Southampton had the most amount of boarding and deaths being double the survived. Cherbourg was the only one with more li
```




```
Southampton had the most amount of boarding and deaths being double the survived. Cherbourg was the only one with more living th
```

Experiment 2: Fare vs. Survival1. `Fare` is a continuous numerical variable.2. A `boxplot` or `violinplot` is a great way to see the distribution of fares for those who survived vs. those who didn't.3. Create one of these plots to compare the `Fare` distribution by `Survived`. What does it tell you?

```
plt.figure(figsize=(10, 6))
#Shows a box plot of who survived by fare
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival(0 = Died, 1 = Survived)')
plt.xlabel('Survived')
plt.ylabel('Fare($)')
plt.show()
print("It shows me that on avarge the people that spent more money were more likley to surieve the titanic")
```



📝 Knowledge Check**Instructions:** Answer the following questions in this markdown cell.1. **What is the primary goal of Exploratory Data Analysis (EDA)?**2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic? (Describe them in terms of class, gender, and age).3. Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't?**

1. The primary goal of EDA is to help data scientists understand the main idea of the data. It also helps find any outliers or missing values.
2. You had the best chance of survival if you were a female child, and also a passenger in class one.
3. It's important to visualize data instead of just a summary because it can show you outliers that you can't see with the mean, as it can also throw off the mean, so visualization can help you see the shape of the graph across groups.